LY Chhaythean

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12 November 2004 (21 years old)

LY Chhaythean | chhaytheanly

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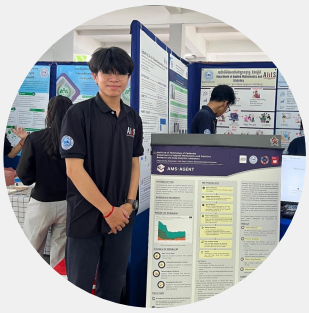

Summary

Fourth-year Data Science & AI student with strong foundations in machine learning, distributed systems, and full-stack development. Passionate about building scalable AI solutions and contributing to open-source projects. Seeking opportunities to apply technical expertise in research or industry settings.

Education

Institute of Technology of Cambodia — *Expected Graduation: August 2027*
Engineer, Data Science, GPA: 4.0 — *Phnom Penh, Cambodia*

Reaksmey Sophorn High School — *Graduated: 2022*
High School Diploma — *Stoung, Kampong Thom*

Experience and Volunteer

Secretary, LLM Basic and Applied Research Club — *June 2025 – Present*
ReDA Laboratory, ITC — *Phnom Penh, Cambodia*

- Contributed to technical research in *Deep Learning*, *Natural Language Processing (NLP)*, and *Image Preprocessing*.
- Studied the mathematical foundations of machine learning models, leading to improved training efficiency and model performance.
- Enhanced the performance of a Gated Recurrent Unit (GRU) model, achieving a 15% increase in accuracy.
- Organized and delivered technical seminars to share research insights with junior members.

Programming Instructor — *August 2025 (Two-week Program)*
Institute of Technology of Cambodia, Kep Campus — *Krong Kep, Cambodia*

- Mentored undergraduate students in programming fundamentals and problem-solving strategies.
- Delivered lectures on Advanced Calculus III and R programming, integrating LaTeX for technical documentation.

Coding and Programming Instructor — *February 2025 – May 2025*
Tech for Kids Academy — *Phnom Penh, Cambodia*

- Volunteered as a mentor, leading the Junior Division at Tech for Kids Academy.
- Taught Scratch programming (developed by MIT) to introduce computational thinking concepts.

Mathematics Tutor (Part-Time) *February 2025 – July 2025*
Start Up Community *Phnom Penh, Cambodia*

- Delivered mathematics training sessions in a private-sector education environment.
- Recognized as an outstanding staff member for teaching effectiveness and student engagement.
- Strengthened time management, communication, and instructional skills.

Mathematics Instructor (Volunteer) *February 2024 – July 2024*
Institute of Technology of Cambodia, Main Campus *Phnom Penh, Cambodia*

- Volunteered as a Mathematics Tutor for undergraduate students.
- Taught Advanced Calculus I & II to first-year engineering students.

PROJECTS

Water-Level Monitoring System *February 2026 – March 2026*
Technologies: Python, Arduino, PHP, SQL

- Designed and developed an intelligent water-level monitoring system integrating IoT sensors with machine learning.
- Implemented real-time leak detection and automated response mechanism to shut off water flow and prevent waste or damage.
- Built end-to-end pipeline including data collection (Arduino), backend processing (Python/PHP), and database management (SQL).
- Developed as a self-initiated project with no budget, demonstrating strong problem-solving and engineering skills.
- Successfully accepted and presented at an AI Conference in Malaysia.
- Paper: *Research Paper*

Khmer Text-to-Speech & OCR System *December 2025 – Februry 2026*
Technologies: Python, FastAPI, HuggingFace Transformers, PyTorch (CUDA), OpenCV

- Designed and implemented an end-to-end Khmer OCR extraction and Text-to-Speech (TTS) pipeline under limited-resource conditions.
- Applied transfer learning on pre-trained vision-language and speech models (HuggingFace) to adapt them for Khmer text recognition and speech synthesis.
- Built advanced image preprocessing techniques (denoising, binarization, geometric correction) to enhance OCR performance on real-world documents.
- Optimized training and inference using PyTorch with CUDA acceleration, reducing inference latency and improving deployment efficiency.
- Developed RESTful APIs using FastAPI for scalable model serving and integration with downstream applications.
- Source code: *Khmer OCR Tesseract*

Website for the Department of Applied Mathematics and Statistics *September 2025 – January 2026*
Tech Stack: NestJS, TypeScript, React, Docker, GitLab CI/CD

- Served as the primary contributor for backend API development using NestJS, designing scalable and maintainable RESTful services.
- Developed a dynamic Admin Dashboard UI with React, enabling real-time content management and role-based access control.
- Implemented DevOps practices, including Docker containerization and GitLab CI/CD pipelines for automated testing and deployment.

- Collaborated with the team to ensure system reliability, performance optimization, and maintainable code architecture.
- Demo: *Website of AMS*

Course Recommendation System *May 2025 – July 2025*

Technologies: Python, Django, Scikit-learn, Pandas, NumPy, SQLite/PostgreSQL

- Designed and developed a full-stack web application for personalized course recommendation using the Coursera dataset.
- Implemented and compared multiple machine learning algorithms, including content-based filtering and collaborative filtering, to generate personalized recommendations.
- Integrated ML models into a Django-based production-ready system with RESTful endpoints and dynamic user interaction.
- Successfully presented the project at the 14th Scientific Day Showcase and was selected as one of the outstanding candidates.
- Strengthened expertise in recommender systems, model evaluation strategies, and deploying ML applications in real-world web environments.
- Source code: *c3ping*

Skills

Programming: Python, TypeScript, SQL, C/C++, Go, LaTeX

ML/AI: PyTorch, TensorFlow, scikit-learn, Hugging Face, LangChain, LLM fine-tuning

Data Analyts: Excel, Matplotlib, Pyplot, R-Programming, Statistics

Web & Systems: React, Django, HTML/CSS, Node.js, NestJS, PostgreSQL, Redis, Docker

Specialized Area: Statistics, Machine Learning, Applied Mathematics, Data Scientist

Tools & Practices: Git, CI/CD, Linux, AWS, MLOps, Testing

Languages: English (*fluent*), Khmer (*native*), French (*intermediate*)

Achievements

Ministry of Education, Youth and Sport (MoEYS) Full Scholarship:

- Awarded a full merit-based scholarship to pursue a Bachelor's Degree in Information Technology, Faculty of Engineering, Royal University of Phnom Penh (RUPP).
- Rejected due to personal reason

Certifications

- 14th Scientific Day
- Senior Teaching Junior 2023-2024
- Academic Performance List

References

1. **Dr. MUTH Boravy**
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